Molecular Featurization I: SMILES and RDKit

Fall 2026

University of South Carolina

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From Lecture 23

Explainable AI

- *Identify influential features using permutation importance*
- *Interpret feature effects using partial dependence plots (PDP)*
- *Explain individual predictions using SHAP analysis*

In this lecture, you will learn

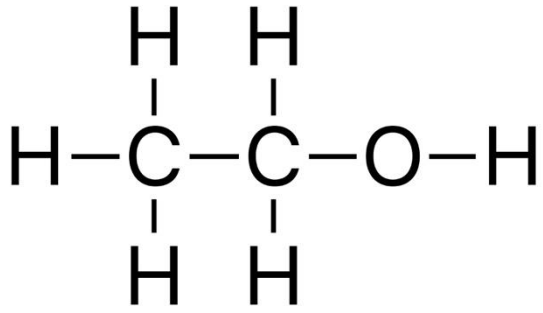
- *Read simple SMILES strings and inspect molecules with RDKit*
- *Represent SMILES tokens using one-hot vectors*
- *Construct and concatenate atom-count and bond-count vectors.*

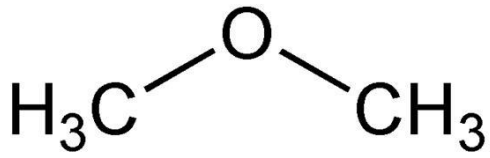
Why Molecular Featurization?



- ML models require numerical inputs.
- Molecular structures contain atoms, bonds, and connectivity.
- Featurization converts chemical information into numerical representations.

Can two different molecular representations lead to different ML predictions?

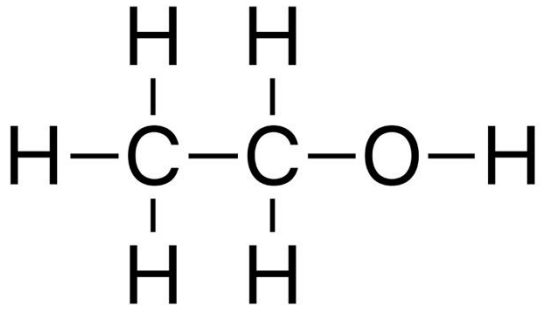




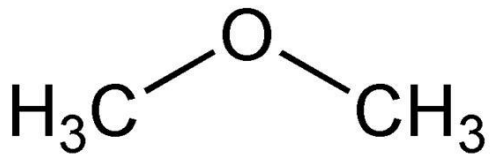
SMILES: Atoms and Bonds

Simplified Molecular Input Line Entry System (SMILES)

SMILES represents molecular structure using a compact text string.



Ethanol
SMILES **CCO**



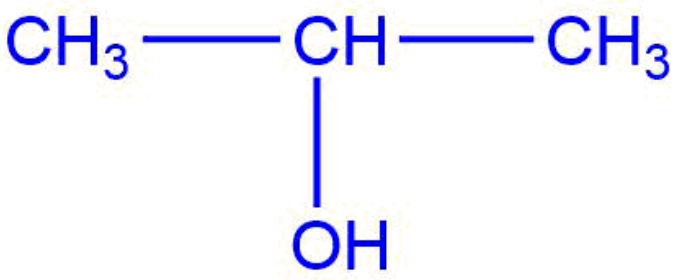
Dimethyl ether
SMILES **COC**

SMILES	Molecule	Reading rule
CC	Ethane	An omitted bond symbol indicates a single bond here
C=C	Ethene	= indicates a double bond
C#C	Ethyne	# indicates a triple bond
CCO	Ethanol	Read the connected atoms in sequence

Many hydrogen atoms are implicit, following normal valence rules

SMILES: Branches

Parentheses introduce a branch from the atom immediately before them.

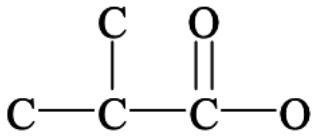


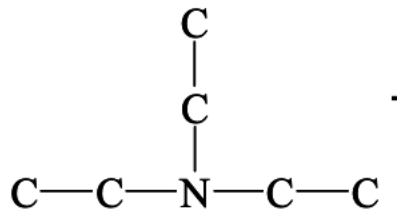
$CH_3–CH(OH)–CH_3$
2-propanol

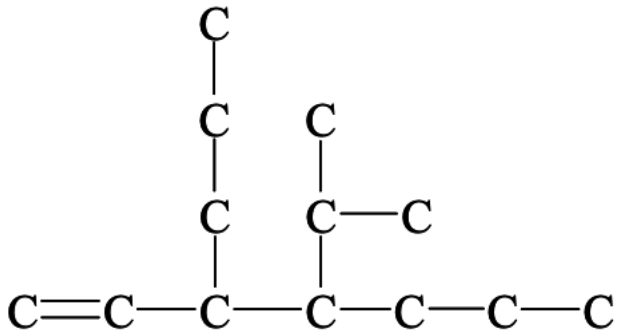
SMILES: **CC(O)C**

- CC: begin the carbon chain.
- (O): attach oxygen to the second carbon.
- Final C: continue from the second carbon.

1. What is **CC(C)C(=O)O**?
2. What is **CCN(CC)CC**?
3. What is **C=CC(CCC)C(C(C)C)CCC**?







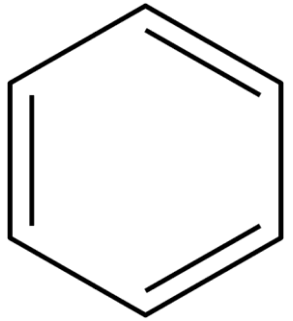
SMILES: Rings and Aromatic Atoms

Ring digits label a connection. Their values do not specify ring size.



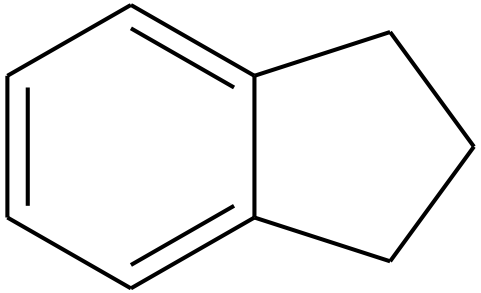
Cyclohexane
SMILES: **C1CCCCC1**

- Matching digits specify a ring closure

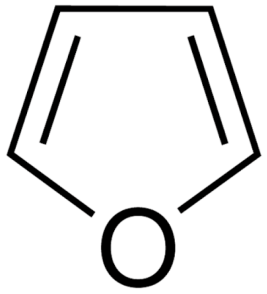


Benzene
SMILES: **c1ccccc1**

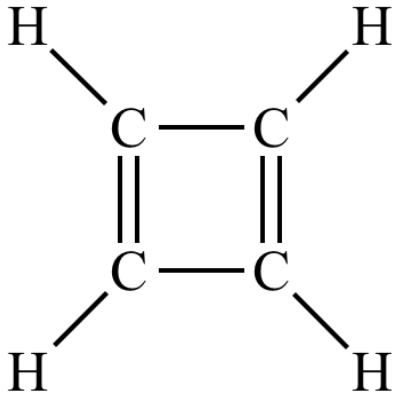
- Lowercase c indicates aromatic carbon



Indane
SMILES: **c1ccc2CCCc2c1**



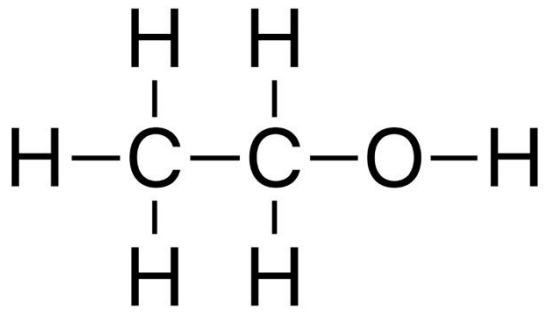
Furan
SMILES: **c1occc1**



Cyclobutadiene
SMILES: **C1=CC=C1**

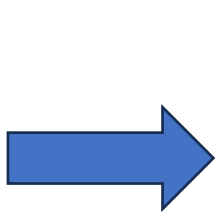
One Molecule Can Have Several SMILES

Different starting atoms and traversal orders can produce different strings.



Ethanol

SMILES **CCO, OCC, C(O)C**





canonicalization



CCO

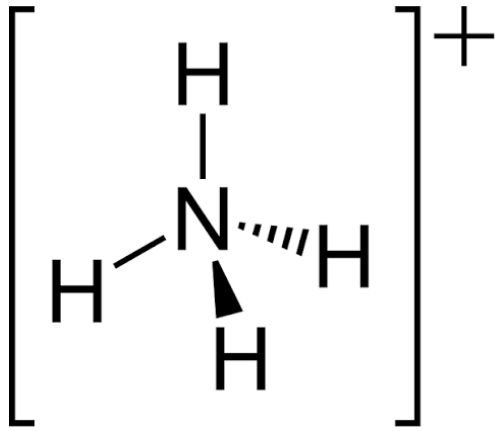
```
In [38]:
core = Chem.MolFromSmiles('O=S(NC1=NOC2=C1C=CC=C2)=O')
core_smiles = Chem.MolToSmiles(core)
print(core_smiles)
core_mol_smiles = Chem.MolFromSmiles(core_smiles)
core_mol_smiles
```

```
O=[SH](=O)Nc1noc2ccccc12
```

Out[38]:



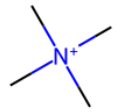
Charges and Stereochemistry



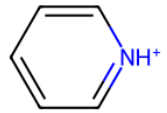
Ammonium ion
[NH4+]



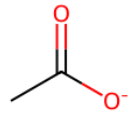
Separate charged components
[Na+].[Cl-]



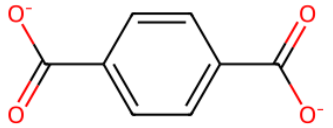
C[N+](C)(C)C



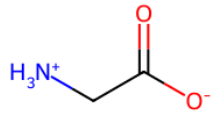
[nH+]1ccccc1



CC(=O)[O-]



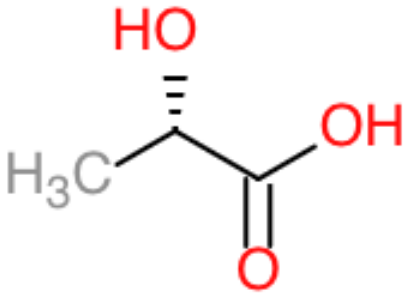
[O-]C(=O)c1ccc(cc1)C(=O)[O-]



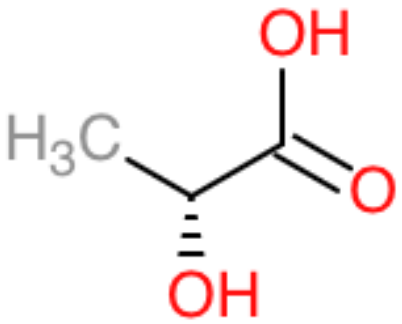
[NH3+]CC(=O)[O-]



Cl[C]Cl



C[C@H](O)C(=O)O



C[C@@H](O)C(=O)O

Lactic acid with specified stereochemistry

SMILES Tokens and One-Hot Encoding

Tokenization divides a SMILES string into an ordered sequence of tokens.

Molecule	SMILES	Token sequence
Ethanol	CCO	[C, C, O]
Dimethyl ether	COC	[C, O, C]

The **vocabulary** contains the distinct token types with the order:

$$V = [C, O]$$

Each token receives a vector with one entry equal to 1 and all other entries equal to 0.

$$C = [1, 0], \qquad O = [0, 1]$$

Position	Ethanol: CCO	Dimethyl ether: COC
1	[1, 0]	[1, 0]
2	[1, 0]	[0, 1]
3	[0, 1]	[1, 0]

Counting Atom Types

A bag-of-words representation records how often each vocabulary token occurs.

Molecule	Number of C tokens	Number of O tokens	Count vector
Ethanol: CCO	2	1	[2, 1]
Dimethyl ether: COC	2	1	[2, 1]

We can obtain the count vector by adding the token's one-hot vectors:

$$[1, 0] + [1, 0] + [0, 1] = [2, 1]$$

Counting retains token frequency but loses sequence order. These two different molecules therefore receive the same representation.

Counting Bond Types

An atom-count representation describes molecular composition.
Bond-type counts add information about which elements connect directly.

Molecule	Structure	C–C single bonds	C–O single bonds
Ethanol	$CH_3–CH_2–OH$	1	1
Dimethyl ether	$CH_3–O–CH_3$	0	2

For the fixed bond vocabulary:

$$V_{\text{bond}} = [\text{C}–\text{C}, \text{C}–\text{O}]$$

$$\mathbf{b}_{\text{ethanol}} = [1, 1], \qquad \mathbf{b}_{\text{ether}} = [0, 2]$$

These molecules have identical atom counts but different bond-type counts.

Combining Atom and Bond Counts

Concatenation places the atom-count and bond-count vectors end to end.

$$\mathbf{x} = [\mathbf{a}, \mathbf{b}]$$

Molecule	C atoms	O atoms	C–C single bonds	C–O single bonds
Ethanol	2	1	1	1
Dimethyl ether	2	1	0	2

$$\mathbf{x}_{\text{ethanol}} = [2, 1, 1, 1]$$

$$\mathbf{x}_{\text{ether}} = [2, 1, 0, 2]$$

The combined vector retains molecular composition and bond-type frequencies.

What Information Is Still Missing?

Different molecules can share both atom counts and bond-type counts.

Molecule	SMILES	Atom counts [C, O]	Bond counts [C–C, C–O]
1-Propanol	CCCO	[3, 1]	[2, 1]
2-Propanol	CC(O)C	[3, 1]	[2, 1]

$$\mathbf{x} = [3, 1, 2, 1]$$

Bond-type counts retain some connectivity information, but they do not specify how all the bonds fit together.

What feature could distinguish an oxygen attached to an end carbon from an oxygen attached to the middle carbon?

Interactive Python Time

24_Molecular_Featurization.ipynb

In this lecture, we learned

- *Read simple SMILES strings and inspect molecules with RDKit*
- *Represent SMILES tokens using one-hot vectors*
- *Construct and concatenate atom-count and bond-count vectors.*

Next lecture:

- *Molecular Fingerprints*
  - *Morgan fingerprints and local atomic environments*
  - *Generating fingerprints with RDKit*
  - *Comparing molecules using fingerprint similarity*